

ASP.net Core Web API

Kafka Integration with C#:

Zookeeper Running -

```
C:\Windows\system32\cmd.exe
[2025-07-09 22:08:49.051] INFO Using org.apache.zookeeper.server.NIOServerCnxnFactory as server connection factory (org.apache.zookeeper.server.ServerCnxnFactory)
[2025-07-09 22:08:49.052] WARN maxCnxns is not configured, using default value 0. (org.apache.zookeeper.server.ServerCnxnFactory)
[2025-07-09 22:08:49.053] INFO Configuring NIO connection handler with 18s sessionless connection timeout, 2 selector thread(s), 32 worker threads, and 64 k B direct buffers. (org.apache.zookeeper.server.NIOServerCnxnFactory)
[2025-07-09 22:08:49.105] INFO binding to port 0.0.0.0/0.0.0.0:2181 (org.apache.zookeeper.server.NIOServerCnxnFactory)
[2025-07-09 22:08:49.119] INFO Using org.apache.zookeeper.server.watch.MatchManager as watch manager (org.apache.zookeeper.server.watch.MatchManagerFactory)
[2025-07-09 22:08:49.120] INFO Using org.apache.zookeeper.server.watch.MatchManager as watch manager (org.apache.zookeeper.server.watch.MatchManagerFactory)
[2025-07-09 22:08:49.128] INFO zookeeper.snapshotSizeFactor = 0.33 (org.apache.zookeeper.server.ZKDatabase)
[2025-07-09 22:08:49.128] INFO zookeeper.commitLogCount=500 (org.apache.zookeeper.server.ZKDatabase)
[2025-07-09 22:08:49.126] INFO zookeeper.snapshot.compression.method = CHECKED (org.apache.zookeeper.server.persistence.SnapStream)
[2025-07-09 22:08:49.126] INFO Snapshotting: 0x0 to \tmp\zookeeper\version-2\snapshot.0 (org.apache.zookeeper.server.persistence.FileTxnSnapLog)
[2025-07-09 22:08:49.130] INFO Snapshot loaded in 38 ms, highest zxid is 0x0, digest is 11770dbb04 (org.apache.zookeeper.server.ZKDatabase)
[2025-07-09 22:08:49.131] INFO Snapshotting: 0x0 to \tmp\zookeeper\version-2\snapshot.0 (org.apache.zookeeper.server.persistence.FileTxnSnapLog)
[2025-07-09 22:08:49.132] INFO Snapshot taken in 1 ms (org.apache.zookeeper.server.ZooKeeperServer)
[2025-07-09 22:08:49.141] INFO zookeeper.request.throttler.shutdownTimeout = 10000 ms (org.apache.zookeeper.server.RequestThrottler)
[2025-07-09 22:08:49.141] INFO PrepRequestProcessor (xid:0) started, reconfigEnabled=false (org.apache.zookeeper.server.PrepRequestProcessor)
[2025-07-09 22:08:49.159] INFO Using checkIntervalMs=60000 maxPerMinute=10000 maxNeverUsedIntervalMs=0 (org.apache.zookeeper.server.ContainerManager)
[2025-07-09 22:08:49.160] INFO ZooKeeper audit is disabled. (org.apache.zookeeper.audit.ZKAuditProvider)
```

Kafka Server Running -

```
C:\Windows\system32\cmd.exe
[2025-07-09 22:16:48.489] INFO [AddPartitionsToTxnSenderThread-0]: Starting (kafka.server.AddPartitionsToTxnManager)
[2025-07-09 22:16:48.490] INFO Creating /brokers/ids/0 (is it secure? false) (kafka.zk.KafkaZkClient)
[2025-07-09 22:16:48.497] INFO Stat of the created znode at /brokers/ids/0 is: 25,28,1783615680045,1783615680045,1,0,0,72060670760370620,214,0,70
(kafka.zk.KafkaZkClient)
[2025-07-09 22:16:48.497] INFO Registered broker 0 at path /brokers/ids/0 with addresses: PLAINTEXT://LAPTOP-FPFV1IB4:9092, zxid (broker epoch): 25 (kafka.zk.KafkaZkClient)
[2025-07-09 22:16:48.489] INFO [ExpirationReaper-0-topic]: Starting (kafka.server.DelayedOperationPurgatory$ExpiredOperationReaper)
[2025-07-09 22:16:48.490] INFO [ExpirationReaper-0-Heartbeat]: Starting (kafka.server.DelayedOperationPurgatory$ExpiredOperationReaper)
[2025-07-09 22:16:48.497] INFO [ExpirationReaper-0-Rebalance]: Starting (kafka.server.DelayedOperationPurgatory$ExpiredOperationReaper)
[2025-07-09 22:16:48.506] INFO Successfully created /controller epoch with initial epoch 0 (kafka.zk.KafkaZkClient)
[2025-07-09 22:16:48.508] INFO [GroupCoordinator 0]: Starting up. (kafka.coordinator.group.GroupCoordinator)
[2025-07-09 22:16:48.517] INFO [GroupCoordinator 0]: Startup complete. (kafka.coordinator.group.GroupCoordinator)
[2025-07-09 22:16:48.528] INFO Feature ZK node created at path: /feature (kafka.server.FinalizedFeatureChangeListener)
[2025-07-09 22:16:48.532] INFO [TransactionCoordinator id=0] Starting up. (kafka.coordinator.transaction.TransactionCoordinator)
[2025-07-09 22:16:48.536] INFO [TransactionCoordinator id=0] Startup complete. (kafka.coordinator.transaction.TransactionCoordinator)
[2025-07-09 22:16:48.536] INFO [TxnMarkerSenderThread-0]: Starting (kafka.coordinator.transaction.TransactionMarkerChannelManager)
[2025-07-09 22:16:48.556] INFO [MetadataCache brokerId=0] Updated cache from existing nodes to latest Features(version=3.7-IV0, finalizedFeatures=[], finalizedFeaturesEpoch=0). (kafka.server.metadata.ZkMetadataCache)
[2025-07-09 22:16:48.589] INFO [ExpirationReaper-0-alterACLs]: Starting (kafka.server.DelayedOperationPurgatory$ExpiredOperationReaper)
[2025-07-09 22:16:48.621] INFO [/config/change-event-process-thread]: Starting (kafka.common.ZkNodeChangeNotificationListener$ChangeEventProcessThread)
[2025-07-09 22:16:48.634] INFO [SocketServer.ListenerTypeZK_BROKER, nodeId=0] Enabling request processing. (kafka.network.SocketServer)
[2025-07-09 22:16:48.639] INFO awaiting socket connections on 0.0.0.0:9092. (kafka.network.DataPlaneAcceptor)
[2025-07-09 22:16:48.648] INFO Kafka version: 3.7.0 (org.apache.kafka.common.utils.AppInfoParser)
[2025-07-09 22:16:48.648] INFO Kafka commitId: 2ae524e0625438c5 (org.apache.kafka.common.utils.AppInfoParser)
[2025-07-09 22:16:48.648] INFO Kafka startTimeMs: 1783615680642 (org.apache.kafka.common.utils.AppInfoParser)
[2025-07-09 22:16:48.652] INFO [KafkaServer id=0] started (kafka.server.KafkaServer)
[2025-07-09 22:16:41.184] INFO [zk-broker-0-to-controller-forwarding-channel-manager]: Recorded new controller, from now on will use node LAPTOP-FPFV1IB4:9092 (id: 0 rack: null) (kafka.server.NodeToControllerRequestThread)
[2025-07-09 22:16:41.205] INFO [zk-broker-0-to-controller-alter-partition-channel-manager]: Recorded new controller, from now on will use node LAPTOP-FPFV1IB4:9092 (id: 0 rack: null) (kafka.server.NodeToControllerRequestThread)
```

Topic Created -

```
C:\Windows\system32\cmd.exe - Command Prompt - librdkafka
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Rohit Pamar>cd..
C:\Users>cd..
C:\>cd kafka_2.13-3.7.0

C:\kafka_2.13-3.7.0>.\bin\windows\kafka-topics.bat --create --topic chat-messages --bootstrap-server localhost:9092 --partitions 1 --replication-factor 1
Created topic chat-messages.

C:\kafka_2.13-3.7.0>.\bin\windows\kafka-topics.bat --list --bootstrap-server localhost:9092
chat-messages

C:\kafka_2.13-3.7.0>|
```

Creating Publisher in Command Prompt -

```
C:\Windows\system32\cmd.exe - Command Prompt - librdkafka
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Rohit Pamar>cd..
C:\Users>cd..
C:\>cd kafka_2.13-3.7.0

C:\kafka_2.13-3.7.0>.\bin\windows\kafka-topics.bat --create --topic chat-messages --bootstrap-server localhost:9092 --partitions 1 --replication-factor 1
Created topic chat-messages.

C:\kafka_2.13-3.7.0>.\bin\windows\kafka-topics.bat --list --bootstrap-server localhost:9092
chat-messages

C:\kafka_2.13-3.7.0>

C:\kafka_2.13-3.7.0>.\bin\windows\kafka-console-producer.bat --topic chat-messages --bootstrap-server localhost:9092
>Hello World
*
```

```
Command Prompt - librdkafka
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Rohit Pamar>cd..
C:\Users>cd..
C:\>cd k
The system cannot find the path specified.

C:\>cd kafka_2.13-3.7.0

C:\kafka_2.13-3.7.0>.\bin\windows\kafka-console-consumer.bat --topic chat-messages --bootstrap-server localhost:9092 --from-beginning
Hello World
|
```

Console Chat App

Output -

```
E:\Cognizant Deep Skill Study\DotNet_FSE_Angular_hands_on_Exercise\Week 3 Solved Exercise\Asp.net Core 8.0 Web API\KafkaChatConsole\bin\Debug\net8.0>KafkaChatConsole.exe
Enter your username: Alice
Connected as 'Alice'. Type a message and press Enter (type 'exit' to quit):

You: Hello Bob
You:
[Alice]: Hello Bob
You:
[Bob]: Hello Alice
You:
[Bob]: how are you ?
You: I am Fine.
You:
[Alice]: I am Fine.
You:

E:\Cognizant Deep Skill Study\DotNet_FSE_Angular_hands_on_Exercise\Week 3 Solved Exercise\Asp.net Core 8.0 Web API\KafkaChatConsole\bin\Debug>cd net8.0

E:\Cognizant Deep Skill Study\DotNet_FSE_Angular_hands_on_Exercise\Week 3 Solved Exercise\Asp.net Core 8.0 Web API\KafkaChatConsole\bin\Debug\net8.0>KafkaChatConsole.exe
Enter your username: Bob
Connected as 'Bob'. Type a message and press Enter (type 'exit' to quit):

You: Hello Alice

[Bob]: Hello Alice
You: You: how are you ?
You:
[Bob]: how are you ?
You:
[Alice]: I am Fine.
You: |
```